

WCA Long Term Forecast for Cement and Clinker Demand



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Introduction

The cement industry faces unprecedented changes over the next 25 years in the course of a transition to a decarbonised future. Many of the factors that will have a large impact by 2050 are still in an early stage and timing of their impact is uncertain. This forecast looks at what that impact might be and challenges the assumptions behind some of the more commonly cited long term forecasts. Key takeaways are:

- Global cement demand in 2050 is likely to be much lower than current forecasts, around 3 billion tonnes per annum (tpa)
- Demand for clinker, the key ingredient contributing to carbon emissions, will decline at an even steeper rate, down to 1.5 billion tpa
- This has significant implications for the scale of cement's unabated carbon emissions by 2050. Our central scenario is for 0.8 million tpa CO₂ before considering the impact of CCS
- This picture is not currently well understood and will inform the demand for CCS projects and reduce the number of cement plants at which CCS is needed.

Background

Earlier this year one of the WCA Members noted that there was no available forecast that looked at the outlook for clinker demand in light of the impact of new technologies on clinker-free cements and the increasing use of supplementary cementitious materials and other clinker substitutes (which I will collectively refer to as SCMs) in cement. This report looks at factors that will impact demand for clinker between now and 2050 and sets out a potential range for projected demand.

The outlook for clinker demand is critical in estimating the need for Carbon Capture and Storage (CCS) installations. CCS projects are capital intensive and raising the money for CCS will be a significant challenge in markets with overcapacity and low prices. Having a realistic view of the likely requirements is therefore critical for policymakers and cement producers alike.

Outlook for Cement Demand

In an excellent report setting out the challenges faced by the cement industry, the Energy Transitions Commission referred to the International Energy Agency's (IEA) 2018 report, which stated that the current global demand of circa 4.2 billion tpa is forecast by the IEA's Reference Technology Scenario to grow to 4.7 billion tonnes by 2050 [1]. This represents an increase of 12% from 2020 levels.

The Global Cement & Concrete Association's underlying assumption in its Concrete Future roadmap is a Business as Usual increase in CO2 emissions from 2.7 billion tpa in 2020 to 3.8 billion in 2050 [2]. (Based on simple linear extrapolation)

For reasons which we will look at later in this article, these forecasts of significant growth are unlikely. Indeed, in its latest forecast the IEA is now projecting flat demand (at 2020 levels) until 2035 followed by a 5% decline by 2050. Our estimates of 2024 cement demand already show a reduction of 9% from 2020 and we expect a further decline of 22% 2050, resulting in a level of 3.0 billion tpa.

'Business As Usual' Projections of Cement Demand

Business as Usual (BAU) here refers to projections before considering new developments that will reduce the demand for conventional, clinker based cements. These developments include the replacement of concrete with other materials such as timber, the use of clinker-free binders, design optimisation, and the reduction of waste in the supply chain.

To assess the likely demand, it is helpful to example four types of markets, as shown in the table below:

[1] ETC (2018), *Mission Possible: Reaching net-zero carbon emissions from harder-to-abate sectors by mid-century*.

[2] GCCA (2021), *The GCCA 2050 Cement and Concrete Industry Roadmap for Net Zero Concrete*.

Table 1: Split of global markets into 4 types

Market Type	Share of world consumption		Regions
	2024e	2035f	
1. Decline then stable: China	48%	32%	China
2. Broadly stable: Developed economies	13%	13%	NE Asia E Europe & Turkey W Europe North America Oceania
3. Slow growth: Middle income regions	22%	26%	CIS Middle East North Africa Latin America SE Asia
4. Fast growth: Emerging markets	18%	28%	Sub Saharan Africa South Asia

1 Demand in China will decline before reaching stability

In 2020 China had a cement demand of 2.4 billion tonnes representing 57% of world consumption, a ration that had remained roughly stable since 2014. This represents a per capita consumption of 1,700 kg per capita per year compared with an average of 300 kg in the rest of the world.

A comparison with other East Asian economies is instructive. In both Japan and Taiwan consumption dropped rapidly from their peaks of 700 and 1,000 kg per capita to a level of about half of this. South Korea is perhaps the closest comparison to China, with a peak of almost 1,400 kg per capita before the 1998 Asian Financial Crisis which dropped to around 850 kg per capita, a level it has maintained since. The drop in China is likely to be more pronounced (as a percentage of peak demand) than in South Korea for two main reasons:

- 1.The Chinese peak was sustained over eight years and the resulting cumulative consumption is significantly higher than Korea's at its peak.
- 2.The Chinese government brought forward many infrastructure projects to prop up demand during the period of slowed GDP growth, bringing forward a portion of demand from the next 5-10 years.

Cement demand in China has fallen almost 25% from 2.4 billion tonnes in 2020 to 1.8 billion tonnes in 2024, a decline of 7% p.a. We expect demand to continue to fall by 3.9% p.a. to 1.2 billion tonnes in 2035 and then at a slower 1.6% p.a. to 0.9 billion tonnes by 2050 reflecting both lower per capita consumption and a decline in population.

2 Developed countries will be broadly stable

This includes the largely developed markets in Europe (including Türkiye), North America, Northeast Asia and Oceania and represents just 13% of world demand for cement. Türkiye and some East European countries are still benefiting from enhanced demand from their development boom but the impact of this is modest and will fade over by 2050.

There is demand growth expected in N. America and Oceania due to population growth, but declines are expected in Northeast Asia and Western Europe due to population decline. In 2024, these developed markets are estimated to have demand just under 0.5 billion tonnes, a level they are expected to maintain over the whole period to 2050.

3 Middle income countries will experience sustained slow growth

In this group of middle-income countries, including the CIS, Middle East, North Africa, Latin America and Southeast Asia, steady growth is expected from 0.84 billion tonnes in 2024 to 0.95 billion in 2035 and 1.03 billion in 2050. This represents an average growth rate of 0.8% p.a. just slightly ahead of population growth which is forecast to be 0.7% p.a.

This is a heterogeneous group with different trends in different countries. Population growth varies considerably as does the stage of development. Some countries, such as Saudi Arabia and the UAE, will reach a stage of development where cement consumption will start to decline. Others, such as the central Asian republics, will continue to experience rapid population and per capita consumption growth due to ongoing development.

4 Emerging Markets will see both population and consumption growth

Emerging Markets in South Asia and Sub-Saharan Africa will see rapid growth in cement demand, almost doubling from 0.69 billion tonnes in 2024 to 1.22 billion tonnes in 2050 while population grows by 40% from 3.1 billion people to 4.4 billion. Per capita consumption will increase from 220 kg per year to 280 kg, almost reaching Western European levels.

For a more detailed (and slightly more optimistic) look at the Indian market through 2035, see Joachim Harder, OneStone Consulting in VDZ October 2024 (link to be provided).

Table 2: BAU projections of Cement Demand

Market Type	Regions	Consumption (Million tpa)			
		2020	2024e	2035f	2050f
1. Decline then stable:	China	2,411	1,825	1,183	928
2. Already stable	North America	115	120	131	138
	W Europe	131	126	129	125
	E Europe & Turkey	120	127	129	98
	Oceania	12	12	13	14
	NE Asia	97	97	89	85
3. Slow growth	SE Asia	236	240	275	297
	North Africa	93	97	111	131
	Latin America	164	176	202	215
	CIS	97	116	127	136
	Middle East	196	208	235	254
4. Fast growth	South Asia	392	537	813	908
	Sub Saharan Africa	130	153	207	308
	Total	4,194	3,834	3,643	3,637

The more granular picture outlined above reveals why current BAU forecasts are likely to be wrong, and WCAs estimate is 3.6 tpa. However, this is not the whole picture, as the pressure to decarbonise means we should not expect BAU norms to continue in the years ahead.

Potential Further Reductions in Demand for Cement

Everyone in the cement industry is aware of the ever-increasing pressure to reduce carbon emissions. Over the last 30 years the industry has cut per ton emissions by about 25%. However, our customers – primarily real estate and infrastructure developers – are also under pressure to cut the carbon footprint of their projects, often at a rate that is faster than the cement industry can satisfy. This will lead them to take actions that will reduce the demand for conventional cement.

There have been several well publicised high-rise buildings that have used timber to replace concrete and steel. No doubt we will see more of this in the future. However, limitations in the supply of sustainable timber mean this will not replace a significant portion of concrete. Allied Market Research published an analysis of the mass timber construction market last year which estimating a growth rate of 6% p.a. over the next 10 years.

There is considerable waste in the concrete supply chain, where cement is ultimately being put to use. In ready mixed concrete supply this involves over-ordering and dumping or returning excess concrete. In site mix supply, this involves using more cement than the mix design requires as well as spillage and dust losses. In addition to the reduction of wastage, there is considerable potential to reduce binder demand with greater use of admixtures and the addition of novel materials such as graphene and carbon nanotubes.

Current design practices have been based on cost considerations. As engineers and architects take into account the embodied carbon, there will be reductions in the amount of concrete used to achieve specific objectives. The potential savings have been estimated as over 20%. However, there are other ways to reduce embodied carbon which may be easier to introduce.

Potential Replacement of Traditional Cement with Clinker-free Alternatives

Many companies are exploring and developing non-clinker binder systems, mainly based on geopolymer concretes with steel slag, mine tailings and other industrial byproducts. These are likely to be accepted initially for non-structural and sub-surface applications. These applications which typically use C30 or standard grade concrete and have low risk/loss in the case of failure are estimated to amount to more than 50% of all concrete used.

Scenarios Considered

Because of the uncertainty over if, and when, the factors above will play a significant role at scale, it is hard to quantify their impact precisely, but it is clear they will reduce demand for traditional cement even further.

We have therefore outlined three potential scenarios:

Slow change - Scenario 1: There is progress on all issues but no step change in the rate of adoption

Central - Scenario 2: the potential for the technologies will be fully realised in the developed world but will still be in process in most countries

Rapid change - Scenario 3: by 2050 the full potential of these carbon saving technologies will be realised

Impact in 2050	Less change Scenario 1	Central Scenario 2	More change Scenario 3
Loss of market to timber	4%	5%	6%
Reduction in supply chain waste	6%	8%	12%
Design optimisation	2%	5%	15%
Novel clinker free cements	2%	8%	15%

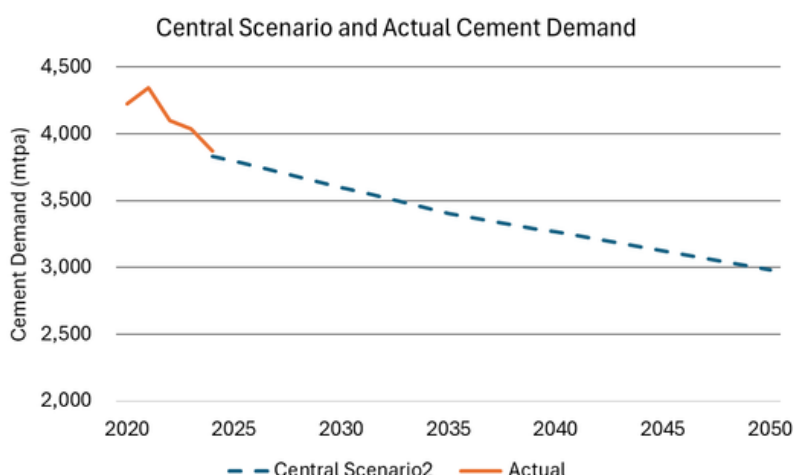
Projections for Cement Demand Including Disruptive Factors

In our view, the cement industry has not widely recognised the forces outlined above and the possible resulting impact on cement demand, which could see a reduction from 4.2 billion tonnes in 2020 to 2.7 billion tonnes in 2050. This would have a major impact on supply/demand balance, pricing and profitability.

Table 3: Projections for cement demand under 3 possible scenarios

			Scenario 1		Scenario 2		Scenario 3	
	2020	2024e	2035f	2050f	2035f	2050f	2035f	2050f
BAU cement demand (mt)	4,194	3,834	3,643	3,637	3,643	3,637	3,643	3,637
Loss of market to timber			36	145	55	182	73	218
Reduction in supply chain waste			73	218	109	291	146	436
Design optimisation			36	73	73	182	146	546
Adjusted cement demand (mt)	4,194	3,834	3,498	3,200	3,407	2,982	3,297	2,437
Clinker-free cements (mt)			25	73	49	291	74	546
Conventional cement (mt)	4,194	3,834	3,473	3,128	3,358	2,691	3,206	1,891

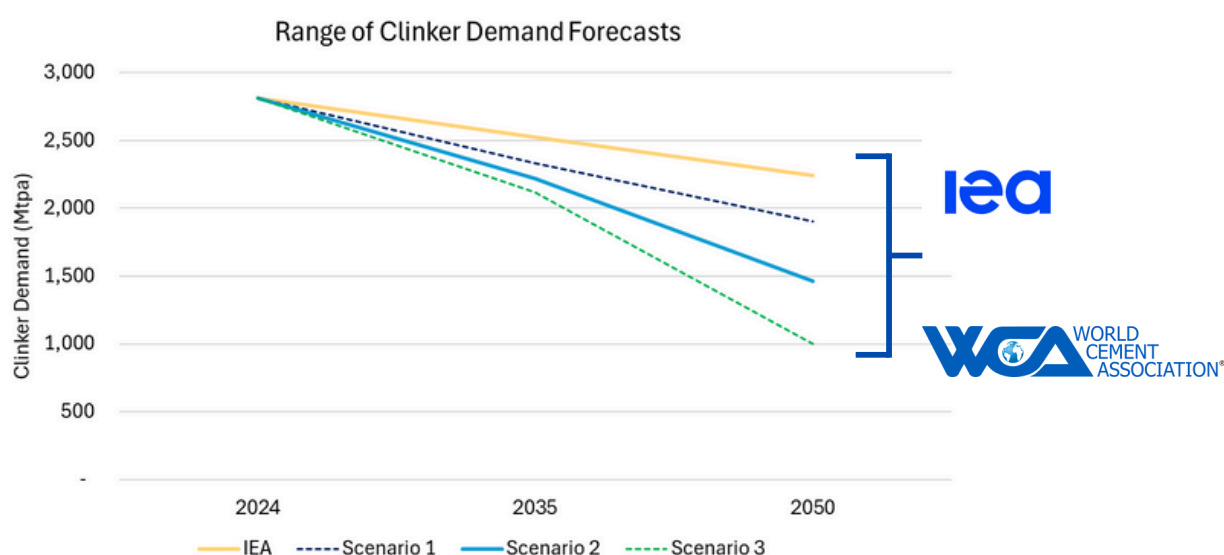
If we look at actual demand since 2020, the trend in cement demand (including clinker-free cements) shows more rapid decline than our Central Scenario 2, as shown in the chart below.



Implications for Clinker Demand

However, cement demand is not the full story as far as the industry is concerned. In addition to the reduction in demand for conventional cement, we can also expect a continued reduction in clinker factor.

Our forecast in the central Scenario 2 is based on a 1% p.a. increase in SCM availability between now and 2035, followed by an increase of 0.5% p.a. from 2035 to 2050. This would result in a reduction in clinker factor from 73% in 2024 to 66% in 2035 and 54% in 2050. For comparison, the IEA forecast is 61% in 2035 and 57% in 2050.



Impact on Carbon Capture and Storage (CCS) Requirements

For the cement industry, the worst option to reduce carbon emissions is CCS. It is expensive and energy intensive. Projects involve a range of parties, making for complex coordination. Furthermore, projects take a long time and often suffer delays. However, we do not have another way of getting to net zero.

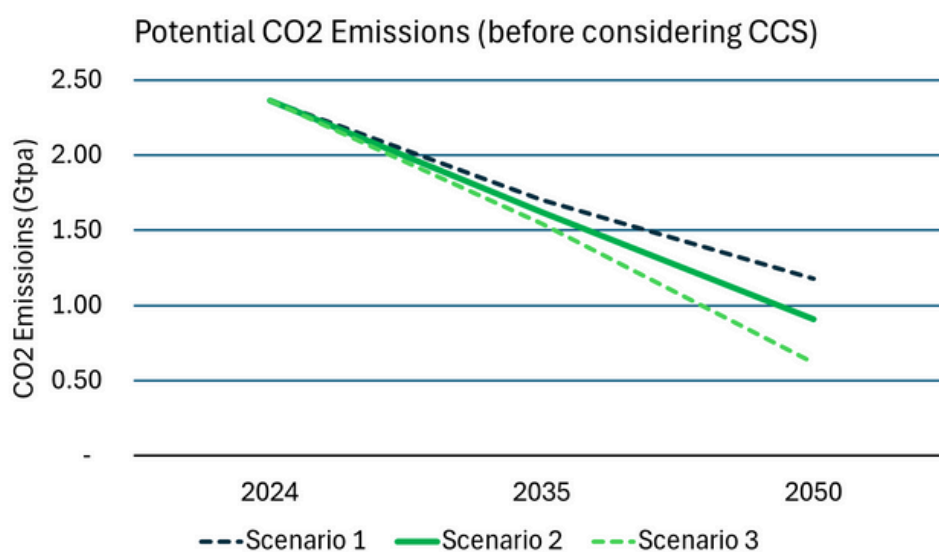
The forecast for clinker demand is therefore critical to calculating what the demand for CCS will be. In 2024 clinker production is projected to be 2.8bn tonnes, a drop of 370 million tonnes from its 2020 peak in just 4 years. This is projected to fall to less than 1.9 billion tonnes by 2050, and perhaps as low as 1.0 billion tonnes.

This has significant implications for demand for CCS equipment and may be sufficient to reduce it below the capital and supply chain limitations that the industry will face as demand for CCS grows rapidly over the next 25 years.

Implications for Cement Industry Carbon Emissions

Clinker production is the main source of cement industry CO₂ emissions. Today, these emissions are roughly one third from burning fuel and two thirds from limestone decomposition. As the cement industry moves from coal to lower carbon fuels such as waste derived fuels and biomass, the fuel emissions per tonne of clinker will be reduced. We expect that by 2050, the industry average specific fuel emissions will have been reduced by nearly 70%.

The projected CO₂ gross emissions from cement production before carbon capture are forecast to fall from 2.4 Gt in 2024 to less than 1 Gt in 2050.



Conclusions

The cement industry faces change on a scale it has not seen in the past and no forecast can be considered definitive. However, it is clear that demand for cement and especially clinker will decline substantially.

The decline of clinker demand to perhaps half of current levels is a huge challenge for the cement industry. Most of the capital invested in cement plants is in the kiln systems to produce clinker. Global capacity utilisation is around 65%. Taking account of expansions that are already being considered, we can expect at least 500 million tpa of new capacity to be added, mostly in emerging markets. By 2050, in scenario 2, kiln utilisation could be just 30%.

These changes also present a big opportunity. Concrete is used in huge quantities because it is an excellent, long-lasting material and because it is very cheap. The future offers the prospect of greater customisation not only to reduce embodied carbon but also to tailor concretes to the application requirements. This is a future in which the agile and forward-looking will prosper.

For more information please contact: events@worldcementassociation.org



Ian Riley, WCA CEO

After graduating from Cambridge University in 1979, Mr Riley worked in a number of management roles in the UK, the USA, Australia and Japan before moving to China in 1998. He was General Manager of a British project management and consulting firm, based in Shanghai and also Chairman of the British Chamber of Commerce in Shanghai before joining Holcim/Huaxin in 2006.

As Huaxin VP responsible for international business, strategy and sustainability, he oversaw Huaxin's first overseas venture in Tajikistan. Mr Riley was Greater China Country Head for LafargeHolcim from 2014 to 2019. He has been CEO of WCA since September 1, 2019.

Who We Are

The World Cement Association, founded in 2016, is the largest independent organisation that works globally on behalf of the cement and clinker industry and its stakeholders, with offices in London and Beijing. We are a not for profit company incorporated under UK law, owned and controlled by our members.

The majority of our members are cement producers, ranging from the world's largest companies, with capacities of 540mt pa, to some of the smallest, with capacities of 0.5mt pa. We also represent companies across the wider value chain, including specialist engineering firms, equipment suppliers and technology providers.

Our members play an active role in the Association through their participation in the Board and four Professional Committees. WCA provides practical help to our members through sharing of best practices in Member Forums, where cement industry leaders in areas as diverse as plant design, green quarries, machine learning AI, vertically integrated business models and low carbon footprints can disseminate their knowledge and experience.

WCA supports a sustainable cement industry, encouraging technical development and innovation by its members to achieve full decarbonisation, in line with the UN's Sustainable Development Goals.



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